Certified Ethical Hacker Study Notes

CEH Networking & Linux Fundamentals

Clean OCR edition with corrected grammar, flowcharts and Linux command screenshots

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Version 3 · A4 PDF

Ethical Use

This PDF converts rough handwritten CEH notes into clean, student-friendly notes. Diagrams from the notebook have been redrawn as readable flowcharts. Practical security assessment must be performed only on owned systems, authorised labs, or with written permission.

README

How to Use This Playbook

Item	Details
Purpose	Quick CEH revision for networking basics, TCP/IP, OSI model, ports, IP addressing, Linux directories and APT troubleshooting.
Best use	Read one chapter at a time, revise the Important boxes, then practise commands only in your own lab machine.
What changed in v3	OCR spelling cleaned, grammar corrected, notebook diagrams converted into flowcharts, and an expanded Linux command section with terminal-style screenshots.
Ethics	Use these notes only for education and authorised testing. Do not scan, probe or test systems without permission.

Safety Scope

Commands in this PDF are for Linux/Kali learning and system maintenance in your own lab. Anything related to assessment, enumeration or network testing requires clear written authorisation.

***Note** - Hinglish hints are included only where they make a concept easier to remember.*

Contents

All headings, definitions and commands were cleaned from the rough notes while keeping the language simple and exam-focused.

Chapter I

Networking Fundamentals

Computer networking is the technology that allows two or more machines to communicate with each other. Communication happens through hardware, software, rules, ports and interfaces.

Simple Hinglish

Networking ka simple idea: devices ek doosre ko data bhejte hain by following fixed rules. Without rules, devices cannot understand each other.

1.1 What is Computer Networking?

Computer networking means connecting computers and other devices so they can share data, resources and services. Examples include file sharing, Internet access, printers, websites and remote login.

1.2 Types of Networks

Type	Full Form	Scope	Simple Meaning / Example
LAN	Local Area Network	Small area	Home, office, school, college lab. Usually high speed and low latency.
MAN	Metropolitan Area Network	City / campus area	Connects multiple LANs. Example: college + canteen + library + lab + hostel.
WAN	Wide Area Network	Country / continent / global	Connects multiple MANs and LANs globally. Example: Internet / WWW.

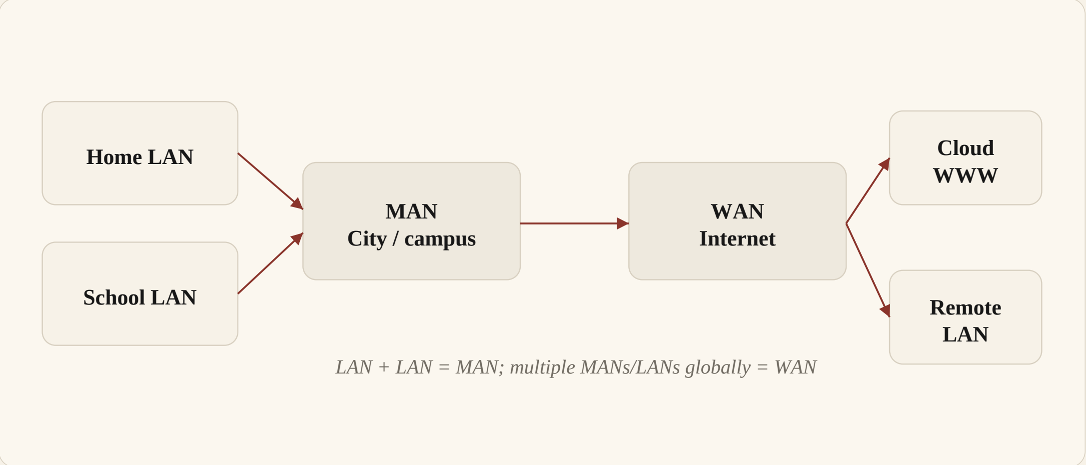

Figure 1. Clean flowchart version of the notebook diagram: LANs combine into MANs, and MANs/LANs connect globally as a WAN.

```
LAN + LAN = MAN
MAN + MAN / LANs across long distance = WAN
Example WAN: Internet (WWW)
```

1.3 Key Networking Entities

The handwritten notes highlighted five important entities that appear repeatedly in networking: IP address, MAC address, ISP, ports and protocols.

Entity	Full Form / Meaning	Student-Friendly Definition
IP Address	Internet Protocol Address	A unique logical address used to identify a device on a network.
MAC Address	Media Access Control Address	A hardware / physical address assigned to a network interface.
ISP	Internet Service Provider	The company that provides Internet connectivity and public IP routing.
Ports	Logical doors	Numbers used by applications to receive and send network data.
Protocols	Rules	A fixed set of rules that defines how communication will happen.

***Note** - Exam tip: IP identifies a device logically on the network. MAC identifies the network interface physically.*

Chapter II

IP Addressing & Network Identity

An IP address gives a device an identity on a network so that data can be delivered to the correct destination. The notes mainly discuss IPv4, public IP, private IP, static IP and dynamic IP.

2.1 IPv4 Address Format

IPv4 is written as four numbers separated by dots. Each number is called an octet. Each octet is 8 bits, so IPv4 is 32 bits in total, which equals 4 bytes.

```
IPv4 example: 17.172.224.47
17    .    172   .    224   .    47
8 bits  8 bits  8 bits  8 bits
Total = 32 bits = 4 bytes
```

Important

Each IPv4 octet is normally from 0 to 255. That is why IPv4 addresses follow a fixed pattern.

2.2 Public IP vs Private IP

Type	Used In	Meaning	Example From Notes
Private IP	LAN	Internal address of a device inside a local network.	`192.168.1.10`
Public IP	WAN / Internet	Internet-facing address used to communicate outside the local network.	`4.4.4.4`

Private IP is used inside a home, lab or office network. Public IP is visible on the Internet side. Routers usually translate private traffic to public traffic when users access the Internet.

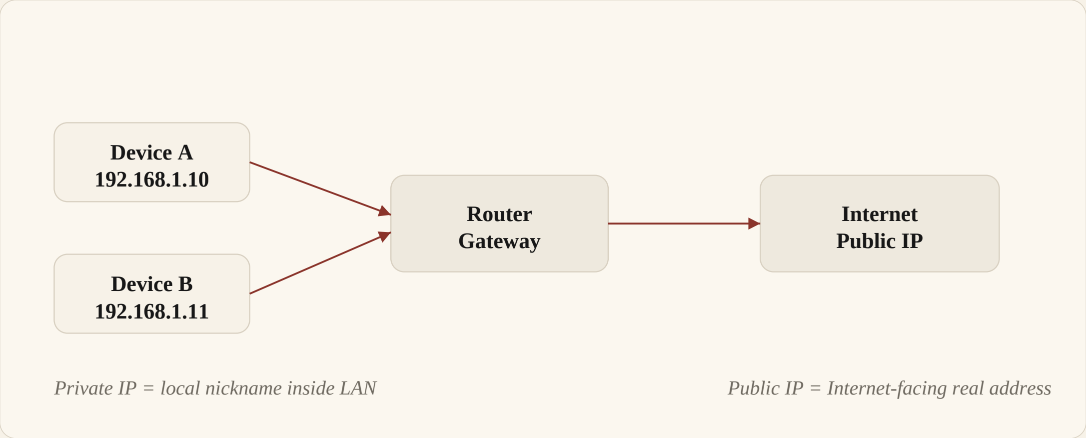

Figure 2. Clean flowchart version of the public/private IP notebook diagram.

***Note** - Hinglish: Private IP ghar ke andar ka nickname hai. Public IP Internet side ka real naam jaisa hota hai.*

2.3 Static IP and Dynamic IP

IP Type	Meaning	Common Use
Static IP	Permanent or manually fixed IP address.	Servers, routers and devices that should always keep the same address.
Dynamic IP	IP address that can change automatically.	Home users and normal client devices. ISP or DHCP can assign it.

Remember

Dynamic IP can change. Static IP remains fixed until it is manually changed or reconfigured.

2.4 Rules of IP Addressing

Devices on the same local network must not use duplicate IP addresses. If two devices use the same IP in one network, communication becomes unreliable because the router/switch cannot clearly know where to send packets.

```
Same LAN example:
A = 192.168.1.10
B = 192.168.1.11

Wrong example:
A = 192.168.1.10
B = 192.168.1.10   # duplicate IP problem
```

***Note** - Local network mein same private IP duplicate nahi hona chahiye.*

2.5 MAC Address

A MAC address is the physical address of a network interface card. It is usually written in hexadecimal pairs separated by colons.

```
MAC address example: 0A:22:84:05:65:04
0A  22  84  05  65  04
8b  8b  8b  8b  8b  8b
Total = 48 bits = 6 bytes
```

Part	Meaning
Vendor / Company ID	The first part identifies the manufacturer or organisation.
Device ID	The remaining part identifies the specific network interface.
Allowed digits	Hexadecimal values: 0-9 and A-F.

***Note** - Common mistake: letters beyond A-F are not valid hexadecimal digits. For example, V is not valid inside a MAC address.*

Chapter III

Ports, Protocols & Data Flow

Ports and protocols are the core of application communication. The notes describe ports as doors of the Internet. A device may have an IP address, but an application still needs a port number to receive data.

3.1 What is a Port?

A port is a logical endpoint for network communication. Think of a system as a building and ports as doors. If the required door is closed, a service cannot receive that traffic.

```
Port number range: 0 - 65535

Simple analogy:
IP address = building address
Port number = specific door inside the building
```

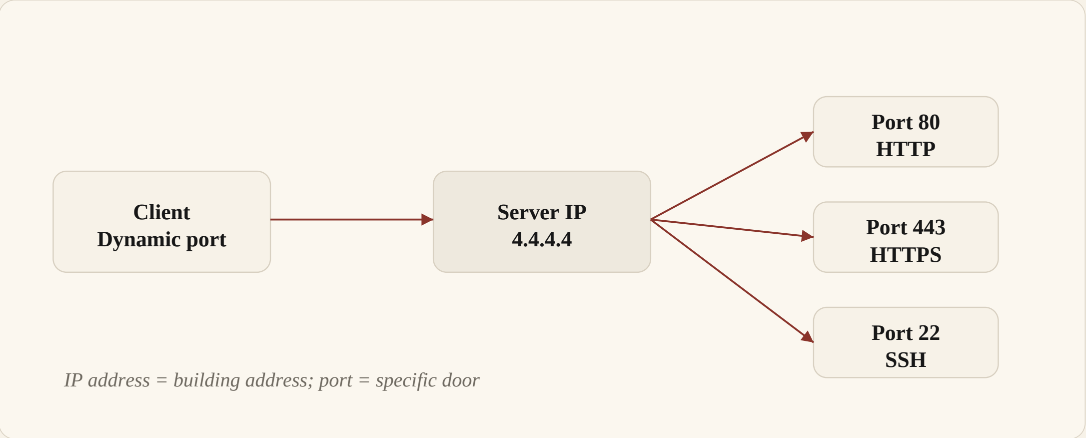

Figure 3. Clean port-flow diagram: the IP address locates the system, and the port selects the service door.

3.2 Types of Ports

Port Range	Category	Meaning
`0 - 1023`	Well-known ports	Used by common services such as HTTP, SSH, DNS and SMTP.
`1024 - 49151`	Registered ports	Assigned to software/applications by vendors or organisations.
`49152 - 65535`	Dynamic / private ports	Temporary client-side ports used during communication.

***Note** - Ports are not bought like IP addresses. They are logical numbers used by operating systems and applications.*

3.3 Default Port Services

Port	Protocol	Application / Service	Simple Use
20	TCP	FTP Data	Transfers file data.
21	TCP	FTP Control	Controls FTP session.
22	TCP	SSH	Secure remote shell access.
25	TCP	SMTP	Sends email.
53	TCP/UDP	DNS	Resolves domain names to IP addresses.
80	TCP	HTTP / WWW	Normal web traffic.
110	TCP	POP3	Receives email from mail server.
443	TCP	HTTPS / TLS	Encrypted web traffic.

Clarification

HTTP normally uses port 80. HTTPS usually uses port 443 because HTTP is protected using TLS/SSL encryption.

3.4 What is a Protocol?

A protocol is a set of rules that defines how data is formatted, transmitted, received and acknowledged. Protocols make communication systematic and predictable.

```
Protocol = rules for data flow
Examples: HTTP, HTTPS, DNS, SSH, TCP, UDP, ARP, DHCP
```

Chapter IV

TCP/IP Model, OSI Model & Transport Protocols

The notes compare the OSI model and the TCP/IP model. OSI is a seven-layer reference model. TCP/IP is the practical Internet model used in real networks.

4.1 OSI Model

Layer No.	OSI Layer	Simple Role
7	Application	User-facing network services such as web, mail and DNS.
6	Presentation	Data formatting, translation, encryption and compression.
5	Session	Starts, maintains and ends communication sessions.
4	Transport	End-to-end delivery using TCP or UDP.
3	Network	Logical addressing and routing using IP.
2	Data Link	MAC addressing and frame delivery on local network.
1	Physical	Cables, signals, radio waves and hardware transmission.

4.2 TCP/IP Model

TCP/IP Layer	Example Protocols	Matches OSI Layers
Application	Telnet, SMTP, POP3, FTP, NTP, HTTP, SNMP, DNS, SSH	OSI 7, 6, 5
Transport	TCP, UDP	OSI 4
Internet	IP, ICMP, ARP, DHCP	OSI 3
Network Access	Ethernet, PPP, ADSL	OSI 2, 1

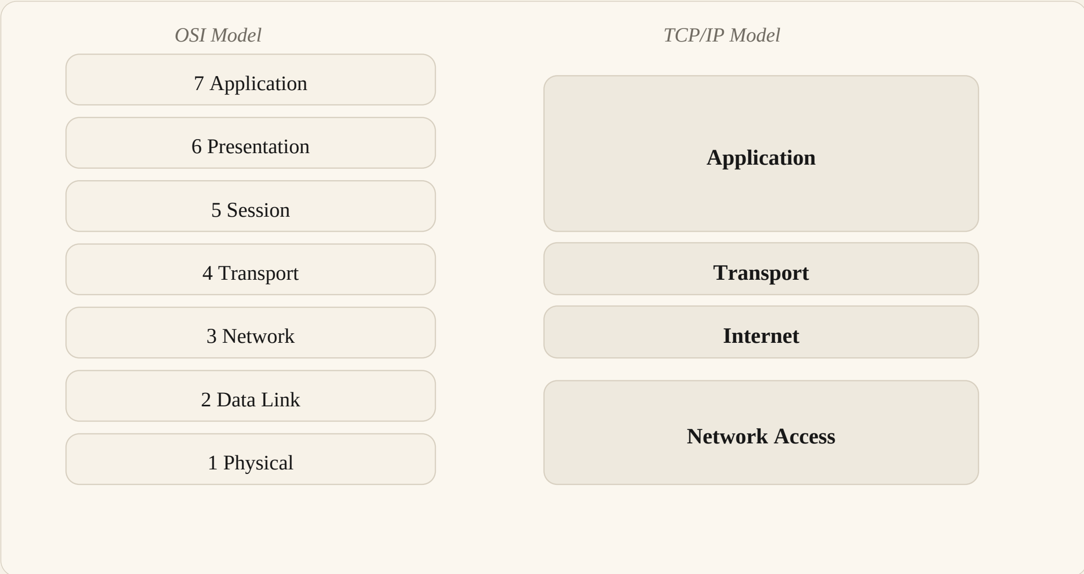

Figure 4. Clean model diagram: OSI seven layers mapped to the four practical TCP/IP layers.

***Note** - TCP/IP is the practical model used in real communication. OSI is mostly used to understand networking concepts layer by layer.*

4.3 TCP vs UDP

Feature	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Reliable; uses acknowledgement and sequencing	No built-in guarantee of delivery
Speed	Slower because of reliability checks	Faster because it has less overhead
Use	When correctness matters	When speed matters

4.4 TCP Three-Way Handshake

TCP creates a reliable connection before data transfer. The notebook diagram showed SYN, SYN+ACK and ACK between two hosts.

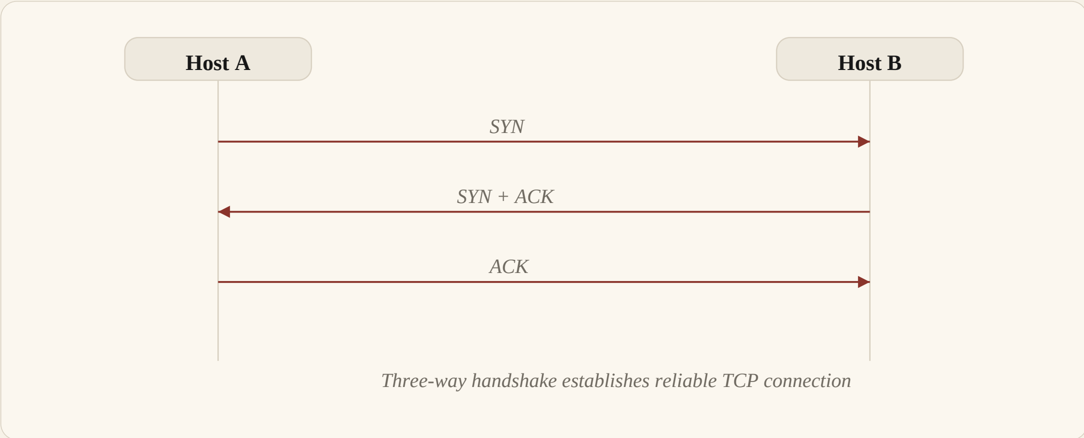

Figure 5. Clean flowchart version of the TCP three-way handshake from the notes.

Flag	Meaning
SYN	Synchronise. Initiates a connection between hosts.
SYN + ACK	Receiver acknowledges SYN and sends its own synchronisation.
ACK	Acknowledgement. Confirms receipt and completes connection setup.

4.5 TCP Session Termination

When communication is finished, TCP closes the session using FIN and ACK messages. FIN means the sender has finished sending data.

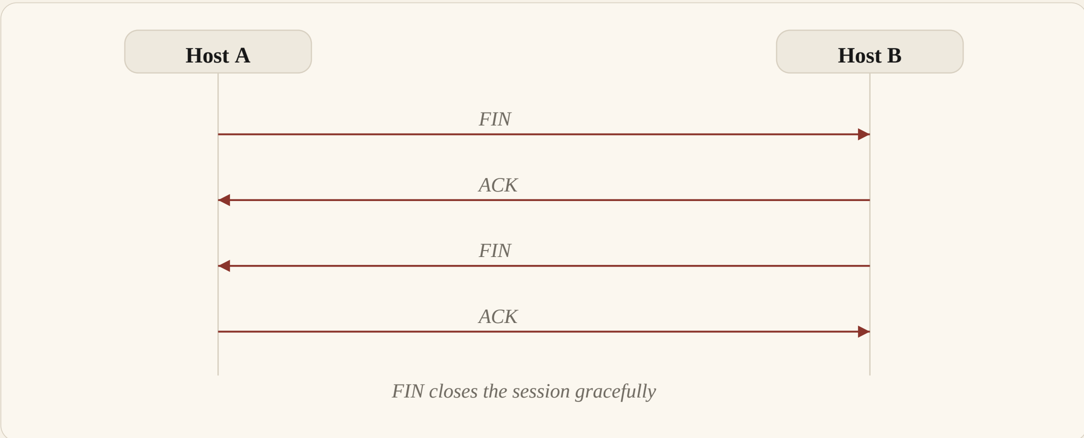

Figure 6. Clean termination flowchart converted from the notebook sequence diagram.

Flag Reminder
ACK acknowledges receipt of a packet. FIN starts graceful termination of the connection.

Chapter V

ARP, DHCP, Localhost & Subnetting

This chapter cleans the notes about automatic IP assignment, address resolution, localhost and subnetting.

5.1 DHCP

DHCP stands for Dynamic Host Configuration Protocol. It automatically gives private IP addresses to users/devices and manages a table of IP and MAC address mappings.

5.2 ARP

ARP stands for Address Resolution Protocol. It helps a device discover the MAC address for a known IP address inside the local network. It commonly works through broadcast request and reply.

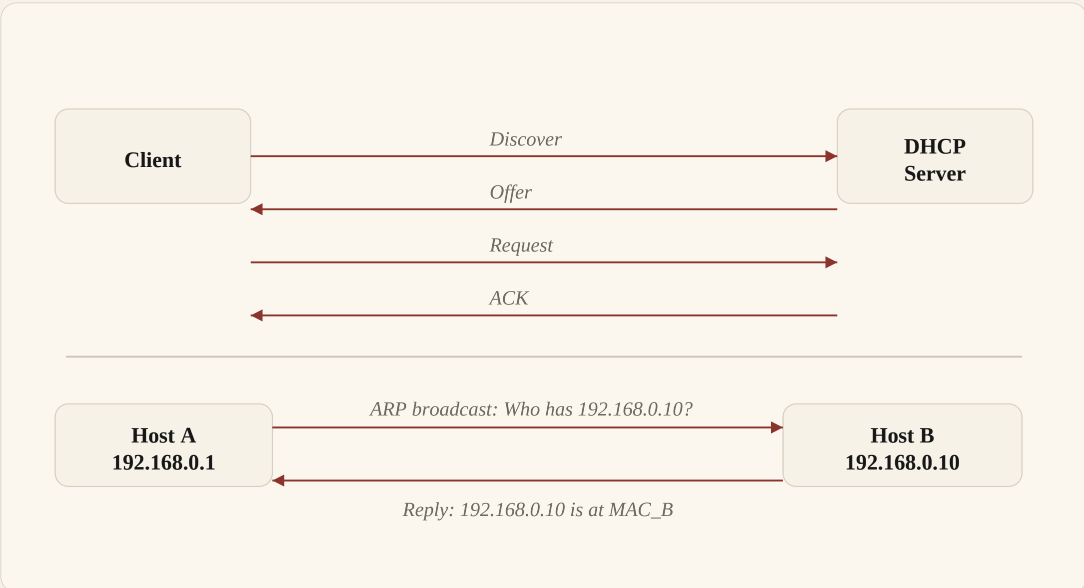

Figure 7. Clean combined flowchart for DHCP assignment and ARP address resolution.

5.3 Localhost

Localhost is the internal IP name of your own system. The common loopback address is 127.0.0.1. It points back to the same machine and is useful for testing local services.

```
localhost = 127.0.0.1
Meaning: this same computer/system
```

***Note** - Hinglish: Localhost apne hi system ka internal address hai. Browser mein local service test karne ke liye use hota hai.*

5.4 Subnetting

Subnetting means dividing one large network into smaller networks. It helps organise IP ranges and control network size.

```
Formula from notes:
Total IPs = 2^(32 - CIDR)

Example:
/24 = 2^(32 - 24) = 2^8 = 256 total IP addresses
```

Subnet Tip

CIDR value jitna bada hota hai, subnet usually utna chhota hota hai. Example: /24 gives 256 total addresses.

Chapter VI

Linux Directory Structure

Linux organises files in a single directory tree starting from root (/). The notebook listed important folders such as /bin, /etc, /lib, /sbin, /root, /usr, /tmp, /boot, /media, /opt and /sys.

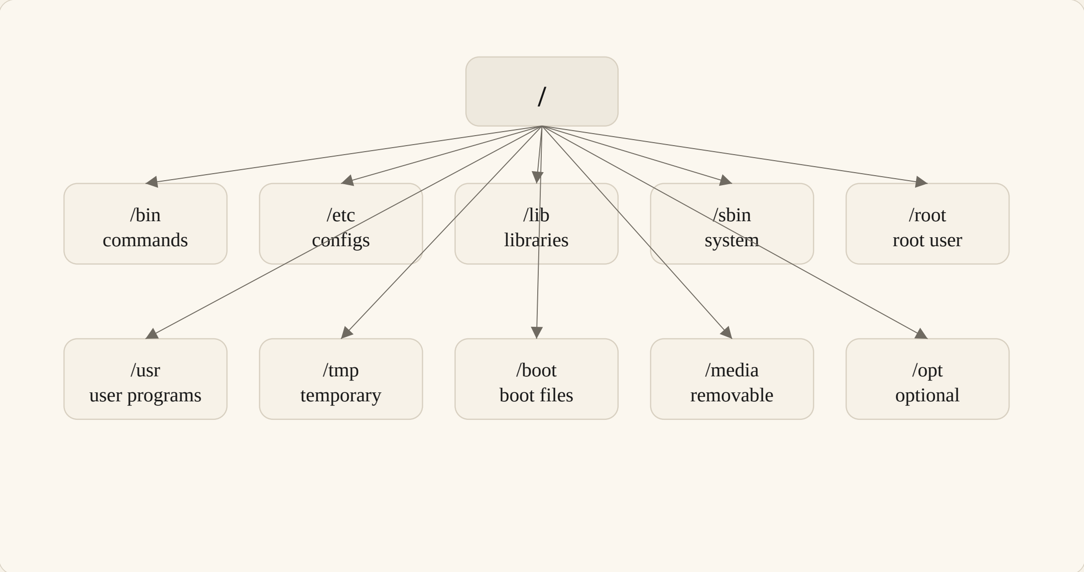

Figure 8. Clean directory-tree diagram converted from the handwritten Linux directory structure notes.

6.1 Important Linux Directories

Directory	Clean Meaning	Examples / Notes
`/bin`	Essential user commands	ls, cp, cat and other basic commands.
`/sbin`	System commands	System administration commands such as iptables and reboot.
`/etc`	Configuration files	passwd, hosts, sshd_config and other config files.
`/root`	Root user home	.bashrc, scripts and .ssh for root user.
`/lib`	System libraries	libc, .so files, modules and kernel-related libraries.
`/usr`	User programs	User-installed programs, shared files and libraries.
`/tmp`	Temporary files	Session temp, cache files and temporary text files.
`/boot`	Boot files	vmlinuz, initrd images and GRUB files.
`/media`	Removable media	USB, CD-ROM and external disks.
`/opt`	Optional software	Custom or third-party apps such as Zoom or Google packages.

Directory	Clean Meaning	Examples / Notes
`/sys`	Kernel/device information	Kernel information, devices, firmware and classes.

Note *- Do not delete system folders manually. Learn their purpose first, then practise only inside a lab VM or safe directory.*

Chapter VII

Linux Commands & APT Troubleshooting

This chapter cleans the Linux command notes. It focuses on help commands, basic file commands, permission symbols and APT troubleshooting.

7.1 Help and Navigation Commands

Command	Meaning
`man`	Open the manual page for a command.
`--help`	Show help/options for a command.
`-h`	Short help option used by many commands.
`pwd`	Print the current working directory.
`ls`	List files and folders.
`ls -l`	Show detailed file listing.

7.2 File and Permission Commands

Command	Meaning
`cp`	Copy files or directories.
`rm -rf`	Remove files/directories recursively and forcefully.
`chmod +x`	Add execute permission to a file.
`chmod -x`	Remove execute permission from a file.

***Note** - Be careful with rm -rf. One wrong path can delete important files. Always verify the path before pressing Enter.*

7.3 Linux Permission Symbols

Linux permissions are commonly shown as r, w and x.

Symbol	Meaning
r	Read permission.
w	Write permission.
x	Execute permission.

```
Example permission style:
rwx rwx rwx
user group others
```

7.4 Updating APT Sources

APT uses source-list files to know where packages should be downloaded from. The notes show editing the sources list and then running an update.

```
cd /etc/apt
mousepad sources.list
apt update
```

Command	What it Does
`cd /etc/apt`	Moves into the APT configuration directory.
`mousepad sources.list`	Opens the package source list in a text editor.
`apt update`	Refreshes package information from configured repositories.

7.5 Cleaning and Rebuilding APT Lists

The notes mention apt clean and manual removal of APT list files. These actions are used when package-list cache becomes messy or outdated.

```
apt clean
apt update

# Another way shown in notes:
rm -rf /var/lib/apt/lists/*
apt update
```

Wildcard Warning

*The asterisk * is a wildcard. It means 'match everything' inside that directory. Use it carefully.*

7.6 Fixing Broken or Missing Package Installation

```
dpkg --configure -a
apt install --fix-broken
apt install --fix-missing
apt reinstall <package_name>
```

Command	Use
`dpkg --configure -a`	Configures packages that were unpacked but not fully configured.
`apt install --fix-broken`	Attempts to fix broken package dependencies.
`apt install --fix-missing`	Tries again when package files are missing from download/cache.
`apt reinstall`	Deletes and installs the selected package again.

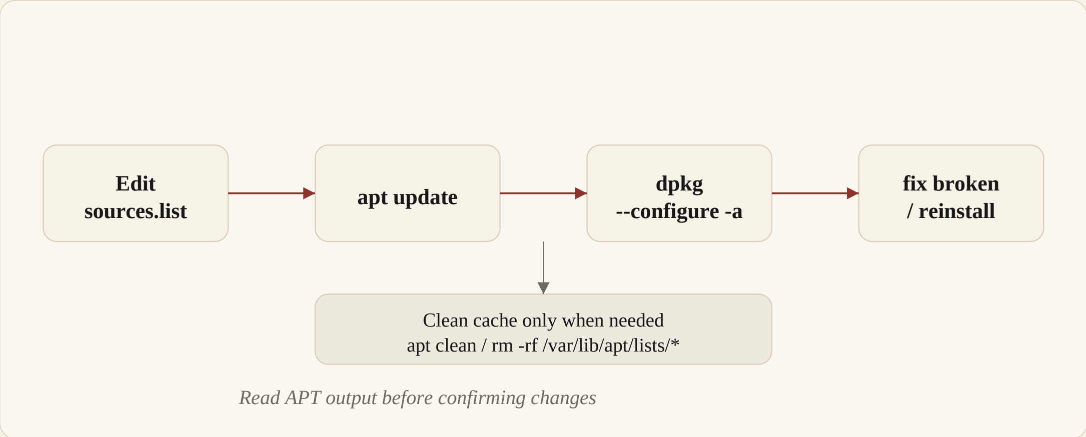

Figure 9. Clean APT troubleshooting flowchart created from the command notes.

7.7 Distribution Upgrade

The notes mention apt dist-upgrade for upgrading the distribution package set. Use it carefully because it can install, remove or change packages to complete an upgrade.

```
apt dist-upgrade
```

***Note** - Kali tip from notes: do not blindly run autoremove/autoclean. Review package changes first because important tools or dependencies may be removed.*

7.8 APT Safety Checklist

Check	Why It Matters
Read the command output before confirming.	APT may install, remove or upgrade many packages at once.
Verify the path before rm -rf.	A wrong path can delete important system files.
Use apt update after changing sources.list.	APT must refresh repository package information.
Prefer targeted reinstall when one tool is broken.	`apt reinstall fixes one package without broad cleanup.`

Practical Order

Troubleshooting order from notes: clean/update the package lists, configure unfinished packages, then fix broken or missing dependencies.

7.9 Essential Linux Command Cheat Sheet

The earlier notebook had only a few Linux commands. This expanded section adds the most important beginner-to-CEH Linux commands with clean terminal-style screenshots. Practise these inside Kali, Ubuntu, Debian or any safe Linux VM.

About Screenshots

Screenshot boxes below are clean recreated terminal examples, not raw photos. They show the command, expected style of output and what you should look for while practising.

Category	Important Commands
Help	man, --help, -h, whatis, apropos
Navigation	pwd, ls, cd, tree, clear, history
Files	touch, cat, less, head, tail, cp, mv, rm, mkdir, rmdir
Search	find, locate, grep, sort, uniq, wc, cut, awk, sed
Permissions	chmod, chown, chgrp, umask, ls -l
Users	whoami, id, groups, passwd, su, sudo -l
Processes	ps, top, htop, kill, pkill, systemctl, journalctl
Networking	ip a, ip route, ping, traceroute, ss, curl, wget, dig, nslookup
System	uname, lsb_release, df, du, free, uptime, date
Packages	apt update, apt install, apt remove, apt search, dpkg, apt reinstall
Archives	tar, gzip, gunzip, zip, unzip

7.10 Help and Manual Commands

Use help commands before running unknown options. This is a safe habit for Linux and CEH labs.

Command	Use
`man ls`	Open the full manual page for ls.
`ls --help`	Show quick option help.
`whatis passwd`	Show one-line meaning of a command.
`apropos network`	Search manual pages related to a keyword.

Help commands

```
rahul@kali:~$ whatis passwd
passwd (1) - change user password
passwd (5) - the password file

rahul@kali:~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about FILEs.
```

Screenshot - Help commands

7.11 Navigation and Listing Commands

Navigation commands help you move around the Linux file system and inspect directories.

Command	Meaning
`pwd`	Print current directory.
`ls`	List files.
`ls -la`	List all files, including hidden files, in long format.
`cd /etc`	Move to /etc directory.
`cd ..`	Go one directory back/up.
`tree`	Show directory structure like a tree.

Navigation and listing

```
rahul@kali:~$ pwd
/home/rahul

rahul@kali:~$ ls -la
total 24
drwxr-xr-x  4 rahul rahul 4096 Jun 13 09:30 .
drwxr-xr-x  3 root  root  4096 Jun 13 09:00 ..
-rw-r--r--  1 rahul rahul  220 Jun 13 09:00 .bashrc
drwxr-xr-x  2 rahul rahul 4096 Jun 13 09:30 notes
```

Screenshot - Navigation and listing

7.12 File Creation, Viewing and Editing

These commands are used daily for reading notes, viewing logs, creating files and making quick edits.

Command	Use
`touch notes.txt`	Create an empty file or update timestamp.
`cat file.txt`	Print full file content.
`less file.txt`	Open file page by page.
`head -n 5 file.txt`	Show first 5 lines.
`tail -n 20 file.txt`	Show last 20 lines.
`nano file.txt`	Edit file in terminal.
`mousepad file.txt`	Edit file in GUI on Kali/Xfce.

Create and read files

```
rahul@kali:~$ touch ceh-notes.txt
rahul@kali:~$ echo 'Networking notes' > ceh-notes.txt
rahul@kali:~$ cat ceh-notes.txt
Networking notes

rahul@kali:~$ head -n 1 ceh-notes.txt
Networking notes
```

Screenshot - Create and read files

7.13 Copy, Move, Delete and Directory Commands

Use these commands carefully because they modify files and folders.

Command	Use
`mkdir labs`	Create a directory.
`rmdir empty_dir`	Remove an empty directory.
`cp file.txt backup.txt`	Copy a file.
`cp -r folder backup_folder`	Copy a directory recursively.
`mv old.txt new.txt`	Rename or move a file.
`rm file.txt`	Delete a file.
`rm -r folder`	Delete a directory recursively.
`rm -rf folder`	Force delete recursively. Dangerous; verify path first.

File operations

```
rahul@kali:~$ mkdir labs
rahul@kali:~$ cp ceh-notes.txt labs/
rahul@kali:~$ mv labs/ceh-notes.txt labs/networking.txt
rahul@kali:~$ ls labs
networking.txt

# Dangerous command - check path first
rahul@kali:~$ rm -rf /path/to/test-folder
```

Screenshot - File operations

Note *- Never practise rm -rf on system folders. Create a test folder first, then practise inside it.*

7.14 Search, Filter and Text Processing

These commands are very useful for log analysis, finding files and extracting useful text from large outputs.

Command	Use
`find /etc -name hosts`	Search by filename.
`locate sshd_config`	Fast filename search using database.
`grep 'error' app.log`	Search text pattern in a file.
`grep -i 'root' /etc/passwd`	Case-insensitive search.
`wc -l file.txt`	Count lines.
`sort file.txt`	Sort lines.
`uniq`	Remove repeated adjacent lines.
`cut -d ':' -f1 /etc/passwd`	Cut field 1 using ':' as delimiter.

Search and filter

```
rahul@kali:~$ grep -i 'ssh' /etc/services | head
ssh             22/tcp
ssh             22/udp

rahul@kali:~$ cut -d ':' -f1 /etc/passwd | head
root
daemon
bin
sys
```

Screenshot - Search and filter

7.15 Permissions and Ownership

Permissions decide who can read, write or execute a file. Ownership decides which user and group own the file.

Command	Use
`ls -l`	View permissions, owner and group.
`chmod +x script.sh`	Add execute permission.
`chmod 755 script.sh`	Owner can rwx, group/others can rx.
`chmod 644 file.txt`	Owner can read/write; group/others can read.
`chown user:group file.txt`	Change owner and group.
`chgrp group file.txt`	Change group only.
`umask`	Show default permission mask.

Permissions screenshot

```
rahul@kali:~$ ls -l script.sh
-rw-r--r-- 1 rahul rahul 44 Jun 13 10:00 script.sh

rahul@kali:~$ chmod +x script.sh
rahul@kali:~$ ls -l script.sh
-rwxr-xr-x 1 rahul rahul 44 Jun 13 10:00 script.sh
```

Screenshot - Permissions screenshot

Permission Reminder

r = read, w = write, x = execute. In rwxr-xr-x, the first group is owner, second is group, third is others.

7.16 User and Privilege Commands

These commands help identify the current user, groups and sudo capabilities.

Command	Use
`whoami`	Show current username.
`id`	Show UID, GID and groups.
`groups`	Show current user's groups.
`passwd`	Change current user's password.
`su -`	Switch to another user/root if credentials are available.
`sudo -l`	List commands the user can run with sudo.

User identity

```
rahul@kali:~$ whoami
rahul

rahul@kali:~$ id
uid=1000(rahul) gid=1000(rahul) groups=1000(rahul),27(sudo)

rahul@kali:~$ groups
rahul sudo
```

Screenshot - User identity

***Note** - Run privilege-related commands only on your own system/lab. Do not attempt privilege abuse on systems you do not own.*

7.17 Process, Services and Logs

Use process and service commands to troubleshoot running programs, services and system errors.

Command	Use
`ps aux`	Show running processes.
`top`	Live process monitor.
`htop`	Interactive process monitor, if installed.
`kill`	Terminate a process by PID.
`pkill`	Terminate processes by name.
`systemctl status ssh`	Check service status.
`systemctl start ssh`	Start a service.
`systemctl stop ssh`	Stop a service.
`journalctl -xe`	Show detailed systemd logs/errors.

Services and processes

```
rahul@kali:~$ ps aux | grep ssh
root       802  0.0  0.1  sshd: /usr/sbin/sshd -D
rahul     950  0.0  0.0  grep --color=auto ssh

rahul@kali:~$ systemctl status ssh
ssh.service - OpenBSD Secure Shell server
     Active: active (running)
```

Screenshot - Services and processes

7.18 Networking Commands

Networking commands help you inspect interfaces, routes, DNS, ports and basic connectivity. Use scanning or probing only in your own lab or with permission.

Command	Use
`ip addr`	Show IP addresses on interfaces.
`ip route`	Show routing table/default gateway.

Command	Use
`hostname -I`	Show assigned IP addresses.
`ping -c 4 8.8.8.8`	Test connectivity with 4 ICMP packets.
`traceroute example.com`	Show route hops, if traceroute is installed.
`ss -tulpen`	Show listening TCP/UDP sockets with process info.
`curl -I https://example.com`	Fetch HTTP response headers.
`wget`	Download a file from a URL.
`dig example.com`	DNS lookup with detailed answer.
`nslookup example.com`	Simple DNS lookup.

Network inspection

```
rahul@kali:~$ ip addr show wlan0
3: wlan0: <BROADCAST,MULTICAST,UP> mtu 1500
    inet 192.168.1.10/24 brd 192.168.1.255 scope global wlan0

rahul@kali:~$ ip route
default via 192.168.1.1 dev wlan0
192.168.1.0/24 dev wlan0 proto kernel scope link
```

Screenshot - Network inspection

Listening ports with ss

```
rahul@kali:~$ ss -tulpen | head
Netid State  Local Address:Port  Peer Address:Port Process
tcp   LISTEN 0.0.0.0:22          0.0.0.0:*      users:((sshd,pid=802))
tcp   LISTEN 127.0.0.1:631       0.0.0.0:*      users:((cupsd,pid=711))
```

Screenshot - Listening ports with ss

Note *- ss is the modern replacement for many netstat checks. It is also a useful command to see which ports are listening on your own machine.*

7.19 System Information, Disk and Memory

These commands help you understand the machine, kernel, storage and RAM status.

Command	Use
`uname -a`	Show kernel and system information.
`lsb_release -a`	Show Linux distribution information, if installed.
`cat /etc/os-release`	Show OS information on most modern Linux systems.
`df -h`	Show disk usage by filesystem.
`du -sh *`	Show size of files/folders in current directory.
`free -h`	Show memory usage.
`uptime`	Show how long the system has been running.
`date`	Show current date and time.

System information

```
rahul@kali:~$ uname -a
Linux kali 6.x-amd64 #1 SMP PREEMPT_DYNAMIC x86_64 GNU/Linux

rahul@kali:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        80G   32G   44G  43% /
```

Screenshot - System information

7.20 Package Management Commands

APT and dpkg commands install, update, remove and repair packages on Debian-based systems such as Kali and Ubuntu.

Command	Use
`apt update`	Refresh package list from repositories.
`apt install`	Install a package.
`apt remove`	Remove package but keep config files.
`apt purge`	Remove package and config files.
`apt search`	Search packages.
`apt show`	Show package details.
`apt reinstall`	Reinstall a broken package.
`dpkg -l`	List installed packages.
`dpkg -i file.deb`	Install a local .deb file.

APT package workflow

```
rahul@kali:~$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
Reading package lists... Done

rahul@kali:~$ apt search nmap
Sorting... Done
Full Text Search... Done
nmap/kali-rolling  Network exploration tool and security scanner
```

Screenshot - APT package workflow

Note *- Install tools only from trusted repositories. Avoid random scripts from the Internet.*

7.21 Archive and Compression Commands

Archives are common when downloading tools, logs, reports or backups.

Command	Use
`tar -xvf file.tar`	Extract a .tar archive.
`tar -xzvf file.tar.gz`	Extract a gzip-compressed tar archive.
`tar -czvf backup.tar.gz folder`	Create compressed tar archive.
`gzip file.txt`	Compress a file to .gz.
`gunzip file.txt.gz`	Decompress a .gz file.
`zip -r backup.zip folder`	Create zip archive recursively.
`unzip file.zip`	Extract zip archive.

Archive extraction

```
rahul@kali:~$ tar -xzvf tool.tar.gz
tool/
tool/README.md
tool/install.sh

rahul@kali:~$ unzip report.zip
Archive:  report.zip
 extracting: report/notes.txt
```

Screenshot - Archive extraction

7.22 Environment, PATH and Shell Helpers

Shell helpers make terminal work faster and help troubleshoot command-not-found issues.

Command	Use
`echo $PATH`	Show command search path.
`export PATH=$PATH:/new/path`	Temporarily add a path.
`source ~/.bashrc`	Reload bash configuration.
`history`	Show command history.
`clear`	Clear terminal screen.

Command	Use
`which`	Show command location.
`file`	Identify file type.
`sha256sum`	Calculate SHA-256 hash.

PATH and file checks

```
rahul@kali:~$ which bash
/usr/bin/bash

rahul@kali:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

rahul@kali:~$ sha256sum notes.txt
2cf24dba5fb0a30e26e83b2ac5b9e29e...  notes.txt
```

Screenshot - PATH and file checks

Revision Point

For CEH labs, these Linux commands are foundational. Strong Linux basics make networking, enumeration and troubleshooting much easier.

Chapter VIII

Information Gathering Basics

Information gathering is the first phase of many security assessments. The handwritten notes divide it into passive and active information gathering.

Ethical Reminder
Only gather information for authorised learning, labs, owned systems or written-permission assessments.

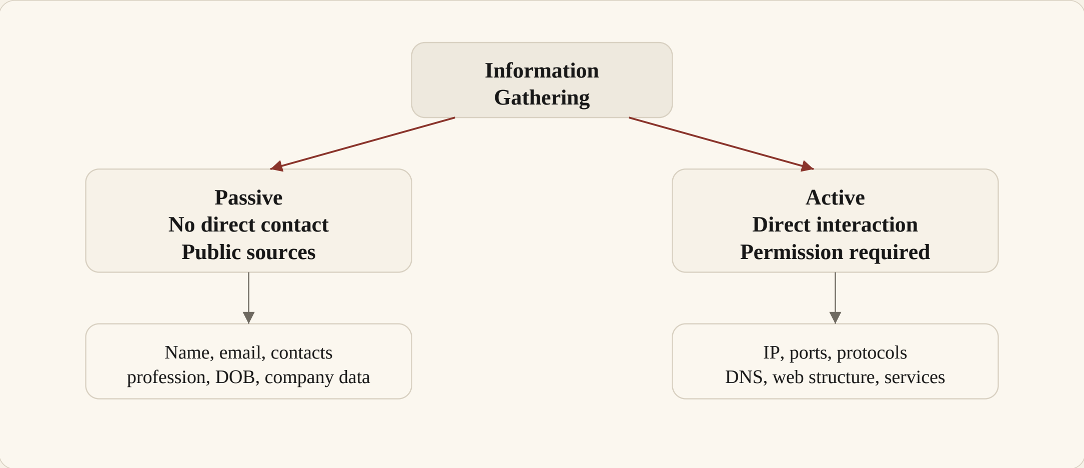

Figure 10. Clean flowchart showing passive vs active information gathering.

8.1 Passive Information Gathering

Passive information gathering means collecting information without directly interacting with the target system. The notes list person-related and company-related data categories.

Category	Examples From Notes
Person-related data	Name, email, contacts, profession, date of birth, address, relations, qualification and public documents.
Company-related data	Company name, performance, technologies, plans/goals, experience, process, daily routine, partners and records.

8.2 Active Information Gathering

Active information gathering means directly interacting with the target network or service to collect technical details. This must only be performed when authorised.

Technical Area	Examples From Notes
Network	IP addresses, domains and DNS records.
Services	Ports, protocols and exposed services.

Technical Area	Examples From Notes
Configuration	Misconfiguration, web structure and technologies used.

***Note** - CEH practical reminder: Passive = observe from public/indirect sources. Active = touch the target system, so permission is mandatory.*

Chapter Appendix

Quick Revision & Command Reference

Use this final section for last-minute revision. It summarises formulas, key definitions and Linux commands from the notes.

A.1 Networking Quick Revision

Term	Quick Meaning
Network	Machines communicating using hardware, software, rules and interfaces.
LAN	Small local network such as home, office, school or lab.
MAN	Large city/campus network connecting multiple LANs.
WAN	Very large network connecting LANs/MANs globally; Internet is the best example.
IP Address	Logical address of a device on a network.
MAC Address	Physical address of a network interface.
Port	Logical door used by applications for communication.
Protocol	Rules for formatting, sending and receiving data.

A.2 Important Numbers

Item	Value
IPv4 size	`32 bits = 4 bytes`
IPv4 octets	`4 octets, each 8 bits`
MAC size	`48 bits = 6 bytes`
Port range	`0 - 65535`
Well-known ports	`0 - 1023`
Registered ports	`1024 - 49151`
Dynamic/private ports	`49152 - 65535`
Loopback IP	`127.0.0.1`
Subnet formula	`Total IPs = 2^(32 - CIDR)`

A.3 Port Reference

Port	Service
`20`	FTP Data

Port	Service
21	FTP Control
22	SSH
25	SMTP
53	DNS
80	HTTP
110	POP3
443	HTTPS / TLS

A.4 Linux Command Reference

```
# Help and navigation
man <command>                 # manual page
<command> --help              # command help
pwd                           # show current directory
ls -la                        # detailed listing with hidden files
cd /path && cd ..             # change directory / go back

# Files and directories
touch file.txt                # create empty file
cat file.txt                  # print file
less file.txt                 # read file page by page
head -n 10 file.txt           # first 10 lines
tail -n 20 file.txt           # last 20 lines
mkdir folder                  # make directory
cp file copy                  # copy file
mv old new                    # move or rename
rm file                       # remove file
rm -rf folder                 # force remove folder - careful

# Search and filter
find /etc -name hosts         # find file by name
grep -i 'ssh' /etc/services  # search text ignoring case
wc -l file.txt                # count lines
sort file.txt | uniq          # sort and remove duplicates
cut -d ':' -f1 /etc/passwd    # cut field by delimiter
```

```
# Permissions and users
ls -l                          # view permissions
chmod +x script.sh             # make executable
chmod 755 script.sh            # rwx for owner, rx for group/others
chown user:group file          # change owner/group
whoami                         # current user
id                             # uid, gid and groups
sudo -l                        # list sudo permissions

# Processes and services
ps aux                         # process list
top                            # live process monitor
kill <PID>                     # kill by PID
systemctl status ssh           # check service
journalctl -xe                 # inspect system logs

# Networking
ip addr                        # show IP addresses
ip route                       # show routes
ping -c 4 8.8.8.8             # test connectivity
ss -tulpen                     # listening TCP/UDP sockets
curl -I https://example.com   # HTTP headers
dig example.com                # DNS lookup
```

A.5 APT Command Reference

```
cd /etc/apt
mousepad sources.list
apt update
apt clean
rm -rf /var/lib/apt/lists/*
dpkg --configure -a
apt install --fix-broken
apt install --fix-missing
apt reinstall <package_name>
apt dist-upgrade
```

For educational and authorised testing purposes only.

Always obtain written permission before conducting security assessments.